

Fast High-Order Compact Exponential Time Differencing Runge–Kutta Methods for Second-Order Semilinear Parabolic Equations

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Received: 23 March 2015 / Revised: 31 August 2015 / Accepted: 3 October 2015 /
Published online: 14 October 2015
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Abstract In this paper we propose fast high-order numerical methods for solving a class of second-order semilinear parabolic equations in regular domains. The proposed methods are explicit in nature, and use exponential time differencing and Runge–Kutta approximations in combination with a linear splitting technique to achieve accurate and stable time integration. A two-step compact difference scheme is employed for spatial discretization to obtain fourth-order accuracy and make use of FFT-based fast calculations. Such methods can be applied to problems with stiff nonlinearities and boundary conditions of Dirichlet or periodic types. Linear stability analysis and various numerical experiments are also presented to demonstrate accuracy and stability of the proposed methods.

Keywords Integrating factor · Exponential time differencing · Linear splitting · Two-step compact difference · Discrete Fourier transforms · Runge–Kutta approximations

Mathematics Subject Classification 65M06 · 65M22 · 65Y20 · 35K58

L. Zhu's research is partially supported by China Fundamental Research of Civil Aircraft under grant number MJ-F-2012-04. L. Ju's research is partially supported by US National Science Foundation under grant numbers DMS-1521965 and DMS-1215659 and by US Department of Energy under grant number DE-SC0008087-ER65393. W. Zhao's research is partially supported by National Natural Science Foundation of China under Grant Number 11171189.

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1 Introduction

Parabolic equations are widely used in many mathematical models for diverse applications, for example, they can represent the scalar diffusion-reaction equations in chemical reactions and population dynamics [11], the Allen–Cahn [1] and Cahn–Hilliard equations [4] for modeling phase transition and separation, the phase field elastic bending energy model for cell membrane simulations [9], the time-dependent advection–diffusion and Navier–Stokes equations for fluids dynamics [32], the Cahn–Hilliard–Navier–Stokes equations for phase-field based multiphase flow [36], the molecular beam epitaxy model for simulating growth of thin films [25, 29], the Ginzburg–Landau equations for modeling superconductivity [8] and so on. In this paper, we consider a family of second-order semilinear parabolic equations with the following form:

$$\frac{\partial u}{\partial t} = D\Delta u - f(u), \quad \mathbf{x} \in \Omega, t \in [t_0, t_0 + T] \quad (1)$$

where Ω is an open rectangular domain in $\mathbb{R}^d$, $T > 0$ is the duration time, $D > 0$ is the diffusion coefficient, and $f(u)$ denotes a potential function. The above Eq. (1) is often accompanied by an initial condition $u|_{t=t_0} = u_0$ for $\mathbf{x} \in \Omega$ and certain boundary conditions. This equation can be viewed as the L^2 gradient flow of a free energy defined by

$$\mathcal{E}(u) = \int_{\Omega} \left\{ \frac{1}{2} |\nabla u|^2 + \frac{1}{D} F(u) \right\} d\mathbf{x}, \quad (2)$$

where $F'(u) = f(u)$ and the energy $\mathcal{E}(u)$ is a decreasing function of time.

Numerical solution of the model equation (1) has attracted much attention in the past decades, especially for problems involving stiff nonlinearities typified by the Allen–Cahn equation [1]. The use of exponential integrators for time discretization has been proven to be quite effective, including methods like the classic *integrating factor* (IF) [17, 18, 23, 26, 27, 34] and the *exponential time differencing* (ETD) [2, 5, 10, 13–16, 21, 28]. They both can maintain the exponential behaviors and differ mainly on the way of evaluating the time integration of the terms resulted from multiplication of the exponential integrator and the function f . The classic IF methods often apply quadrature rules directly to the whole integrand, while the ETD based methods instead first approximate the nonlinear part using polynomial interpolations and then perform exact integrations on the produced new integrands. Thus the ETD methods are usually more accurate than the IF ones due to highly different decaying rates of the exponential integrator components. In [20] an efficient and stable exponential integrator method based on the ETD approach was proposed for solving the Eq. (1), that uses *compact* representation of the central difference scheme for spatial discretization that leads to FFT-based fast calculations, multistep approximations of boundary and stiff nonlinear terms for time integration, and a linear splitting scheme for improving numerical stability. Such a method is then further generalized to the case of Cahn–Hilliard equation, and successfully used for computing material coarsening rates under constant and variable mobilities in [19].

In this paper, we will develop fast high-order ETD Runge–Kutta methods for solving the model equation (1) with a boundary condition of Dirichlet or periodic type. Our methods combine the ETD and Runge–Kutta approximations for accurate time integration with a special two-step compact difference scheme [3, 31] for fourth-order accurate spatial discretization. We specially note that the use of the two-step compact difference scheme allows us to make use of the FFT-based fast calculation techniques (inherited from the central difference scheme) to maintain computational efficiency as that in [19, 20]. Although the classic five-point stencil for Laplacian operator in one dimension is also fourth-order accurate, how-

ever, that approach will impose a difficulty in dealing with points near the boundary and cause troubles in using FFT-based fast calculations.

The rest of the paper is organized as follows. In Sect. 2, we first derive fast and stabilized compact ETD Runge–Kutta schemes (cETDRK) with fourth-order spatial approximations. In Sect. 3, we then present linear stability analyses of the proposed methods. Extensive experiments are carried out in Sect. 4 to numerically demonstrate accuracy and stability of the proposed methods. Finally, some conclusions are given in Sect. 5.

2 Stabilized High-Order Compact ETD Runge–Kutta Methods and Fast Implementation

To deal with problems involving stiff nonlinearities, we first adopt a splitting idea that helps to improve the stability in time integration. The splitting technique used here is analogous to the linear splitting approach used in the stabilized semi-implicit methods for energy stabilization [10, 35, 36], i.e., we rewrite (1) as

$$\frac{\partial u}{\partial t} = (D\Delta - \kappa)u - \hat{f}(u), \quad (3)$$

where $\hat{f}(u) = f(u) - \kappa u$ with a nonnegative constant κ , and it is required that at least

$$\kappa \geq \frac{1}{2} \max_u \left\{ 0, \max_u f'(u) \right\}. \quad (4)$$

Thus we also assume that $f'(u)$ has an upper bound on Ω in this paper; for the problems in which $f'(u)$ has no maximum value, the proposed methods may not work well due to violation of (4).

Without loss of generality we consider the problem (1) in two dimensional space. Suppose $\Omega = \{x_b < x < x_e, y_b < y < y_e\}$. Let us discretize the spatial domain Ω by a rectangular mesh which is uniform in each direction as follows: $(x_i, y_j) = (x_b + ih_x, y_b + jh_y)$ for $0 \leq i \leq N_x$ and $0 \leq j \leq N_y$ with $h_x = (x_e - x_b)/N_x$ and $h_y = (y_e - y_b)/N_y$. Let $u_{i,j} = u_{i,j}(t) \approx u(t, x_i, y_j)$ for $0 \leq i \leq N_x$ and $0 \leq j \leq N_y$ denote the numerical solution. We discretize the time period as $t_n = t_0 + \sum_{s=1}^n \Delta t_{s-1}$, $n = 1, \dots, N_t$ with $t_N = t_0 + T$. We first consider the problem imposed with a Dirichlet boundary condition

$$u = g, \quad (x, y) \in \partial\Omega, t \in [t_0, t_0 + T]. \quad (5)$$

2.1 Two-Step Compact Difference Scheme for Spatial Discretization

By the Taylor expansion, it is easy to check that the following equality holds:

$$\Delta u \Big|_{i,j} = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \Big|_{i,j} = \delta_x^2 u_{i,j} + \delta_y^2 u_{i,j} - \frac{1}{12} \left(h_x^2 \frac{\partial^4 u}{\partial x^4} + h_y^2 \frac{\partial^4 u}{\partial y^4} \right) \Big|_{i,j} + O(h^4), \quad (6)$$

where $h = \max(h_x, h_y)$, and $\delta_x^2 u_{i,j}$ and $\delta_y^2 u_{i,j}$ are the standard central difference discretizations in x, y -directions, respectively, defined by

$$\delta_x^2 u_{i,j} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h_x^2}, \quad \delta_y^2 u_{i,j} = \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h_y^2}.$$

According to (1), we have

$$\begin{aligned}\frac{\partial^4 u}{\partial x^4} \Big|_{i,j} &= \frac{1}{D} \left(\frac{\partial^2}{\partial x^2} \frac{\partial u}{\partial t} - D \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^2}{\partial x^2} f(u) \right)_{i,j} \\ &= \frac{1}{D} \left(\delta_x^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \delta_x^2 (f(u))_{i,j} \right) + O(h^2), \\ \frac{\partial^4 u}{\partial y^4} \Big|_{i,j} &= \frac{1}{D} \left(\frac{\partial^2}{\partial y^2} \frac{\partial u}{\partial t} - D \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^2}{\partial y^2} f(u) \right)_{i,j} \\ &= \frac{1}{D} \left(\delta_y^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \delta_y^2 (f(u))_{i,j} \right) + O(h^2).\end{aligned}$$

Inserting the above two equations into (6), we obtain

$$\begin{aligned}D \Delta u \Big|_{i,j} &= D \delta_x^2 u_{i,j} + D \delta_y^2 u_{i,j} - \frac{h_x^2}{12} \left(\delta_x^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \delta_x^2 (f(u))_{i,j} \right) \\ &\quad - \frac{h_y^2}{12} \left(\delta_y^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \delta_y^2 (f(u))_{i,j} \right) + O(h^4).\end{aligned}\quad (7)$$

Combining (3) and (7), we deduce

$$\begin{aligned}\frac{\partial u}{\partial t} \Big|_{i,j} &= D \delta_x^2 u_{i,j} + D \delta_y^2 u_{i,j} - \frac{h_x^2}{12} \left(\delta_x^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \kappa \delta_x^2 u_{i,j} + \delta_x^2 (\hat{f}(u))_{i,j} \right) \\ &\quad - \frac{h_y^2}{12} \left(\delta_y^2 \frac{\partial u}{\partial t} \Big|_{i,j} - D \delta_x^2 \delta_y^2 u_{i,j} + \kappa \delta_y^2 u_{i,j} + \delta_y^2 (\hat{f}(u))_{i,j} \right) - \kappa u_{i,j} \\ &\quad - (\hat{f}(u))_{i,j} + O(h^4),\end{aligned}$$

that gives us a semi-discrete approximation of the Eq. (3) with fourth-order accuracy in space as

$$\begin{aligned}\left(1 + \frac{h_x^2}{12} \delta_x^2 + \frac{h_y^2}{12} \delta_y^2 \right) \frac{\partial u}{\partial t} \Big|_{i,j} &= \left[D \left(\delta_x^2 + \delta_y^2 + \frac{h_x^2 + h_y^2}{12} \delta_x^2 \delta_y^2 \right) - \left(1 + \frac{h_x^2}{12} \delta_x^2 + \frac{h_y^2}{12} \delta_y^2 \right) \kappa \right] u_{i,j} \\ &\quad - \left(1 + \frac{h_x^2}{12} \delta_x^2 + \frac{h_y^2}{12} \delta_y^2 \right) (\hat{f}(u))_{i,j},\end{aligned}\quad (8)$$

for $1 \leq i \leq N_x - 1$ and $1 \leq j \leq N_y - 1$.

In this case of Dirichlet boundary condition (5), the set of unknowns can be denoted as

$$\mathbf{U} = (u_{i,j})_{(N_x-1) \times (N_y-1)} = \begin{pmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,N_y-1} \\ u_{2,1} & u_{2,2} & \cdots & u_{2,N_y-1} \\ \vdots & \vdots & \ddots & \vdots \\ u_{N_x-1,1} & u_{N_x-1,2} & \cdots & u_{N_x-1,N_y-1} \end{pmatrix}_{(N_x-1) \times (N_y-1)}.$$

Define

$$\mathbf{G}_N^D = \begin{pmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \cdots & 0 & 1 & -2 & 1 \\ 0 & \cdots & 0 & 0 & 1 & -2 \end{pmatrix}_{(N-1) \times (N-1)}.$$

and set

$$\mathbf{A} = \frac{1}{h_x^2} \mathbf{G}_{N_x}^D, \quad \mathbf{B} = \frac{1}{h_y^2} \mathbf{G}_{N_y}^D.$$

Next define the special operators $\mathbb{X}$ and $\mathbb{Y}$ as in [20]:

$$(\mathbf{A}\mathbb{X}\mathbf{U})_{i,j} = \sum_{l=1}^{N_x-1} (\mathbf{A})_{i,l} u_{l,j}, \quad (\mathbf{B}\mathbb{Y}\mathbf{U})_{i,j} = \sum_{l=1}^{N_y-1} (\mathbf{B})_{j,l} u_{i,l},$$

then we can write the semi-discretization of (8) as the following compact representation:

$$\begin{aligned} \left(1 + \frac{h_x^2 \mathbf{A}\mathbb{X} + h_y^2 \mathbf{B}\mathbb{Y}}{12}\right) \frac{d\mathbf{U}}{dt} &= \left[D(\mathbf{A}\mathbb{X} + \mathbf{B}\mathbb{Y}) + \frac{h_x^2 + h_y^2}{12} \mathbf{A}\mathbb{X} \mathbf{B}\mathbb{Y} \right] \mathbf{U} \\ &\quad - \kappa \left(1 + \frac{h_x^2 \mathbf{A}\mathbb{X} + h_y^2 \mathbf{B}\mathbb{Y}}{12}\right) \mathbf{U} \\ &\quad - \left(1 + \frac{h_x^2 \mathbf{A}\mathbb{X} + h_y^2 \mathbf{B}\mathbb{Y}}{12}\right) \mathcal{F}(\mathbf{U}) + \mathcal{W}, \end{aligned} \quad (9)$$

where $\mathcal{F}(\mathbf{U}) = (\hat{f}_{i,j})$ with $\hat{f}_{i,j} = \hat{f}(u_{i,j}) = f(u_{i,j}) - \kappa u_{i,j}$, and $\mathcal{W} = \mathcal{W}^x + \mathcal{W}^y + \mathcal{W}^{xy}$ with $\mathcal{W}^x = (w_{i,j}^x)$, $\mathcal{W}^y = (w_{i,j}^y)$ and $\mathcal{W}^{xy} = (w_{i,j}^{xy})$ defined by

$$w_{i,j}^x = \begin{cases} \frac{D}{h_x^2} g_{0,j} - \frac{1}{12} \frac{\partial g}{\partial t} \Big|_{0,j} - \frac{1}{12} f(g_{0,j}), & i = 1, 1 \leq j \leq N_y - 1, \\ \frac{D}{h_x^2} g_{N_x,j} - \frac{1}{12} \frac{\partial g}{\partial t} \Big|_{N_x,j} - \frac{1}{12} f(g_{N_x,j}), & i = N_x - 1, 1 \leq j \leq N_y - 1, \\ 0, & \text{otherwise,} \end{cases}$$

$$w_{i,j}^y = \begin{cases} \frac{D}{h_y^2} g_{i,0} - \frac{1}{12} \frac{\partial g}{\partial t} \Big|_{i,0} - \frac{1}{12} f(g_{i,0}), & j = 1, 1 \leq i \leq N_x - 1, \\ \frac{D}{h_y^2} g_{i,N_y} - \frac{1}{12} \frac{\partial g}{\partial t} \Big|_{i,N_y} - \frac{1}{12} f(g_{i,N_y}), & j = N_y - 1, 1 \leq i \leq N_x - 1, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$w_{i,j}^{xy} = \frac{D(h_x^2 + h_y^2)}{12h_x^2 h_y^2} \cdot \begin{cases} g_{0,0} + g_{2,0} + g_{0,2} - 2g_{1,0} - 2g_{0,1}, & i = 1, j = 1, \\ g_{N_x,0} + g_{N_x-2,0} + g_{N_x,2} \\ \quad - 2g_{N_x-1,0} - 2g_{N_x,1}, & i = N_x - 1, j = 1, \\ g_{0,N_y} + g_{0,N_y-2} + g_{2,N_y} \\ \quad - 2g_{0,N_y-1} - 2g_{1,N_y}, & i = 1, j = N_y - 1, \\ g_{N_x,N_y} + g_{N_x-2,N_y} + g_{N_x,N_y-2} \\ \quad - 2g_{N_x,N_y-1} - 2g_{N_x-1,N_y}, & i = N_x - 1, j = N_y - 1, \\ g_{i-1,0} - 2g_{i,0} + g_{i+1,0}, & j = 1, 1 < i < N_x - 1, \\ g_{i-1,N_y} - 2g_{i,N_y} + g_{i+1,N_y}, & j = N_y - 1, 1 < i < N_x - 1, \\ g_{0,j-1} - 2g_{0,j} + g_{0,j+1}, & i = 1, 1 < j < N_y - 1, \\ g_{N_x,j-1} - 2g_{N_x,j} + g_{N_x,j+1}, & i = N_x - 1, 1 < j < N_y - 1, \\ 0, & \text{otherwise.} \end{cases}$$

2.2 Exponential Time Integration and Fast Calculation

As shown in [20, 24], **A** and **B** are diagonalizable and have the following eigenvalue decompositions:

$$\mathbf{A} = \mathbf{P}_x \tilde{\mathbf{D}}_x \mathbf{P}_x^{-1}, \quad \mathbf{B} = \mathbf{P}_y \tilde{\mathbf{D}}_y \mathbf{P}_y^{-1},$$

where $\mathbf{P}_x$ and $\mathbf{P}_y$ are orthogonal matrices, $\tilde{\mathbf{D}}_x = \text{diag}[d_1^x, \dots, d_{N_x-1}^x]$ and $\tilde{\mathbf{D}}_y = \text{diag}[d_1^y, \dots, d_{N_y-1}^y]$ with

$$d_k^x = -\frac{4}{h_x^2} \sin^2 \left(\frac{k\pi}{2N_x} \right), \quad k = 1, \dots, N_x - 1, \quad (\mathbf{P}_x)_{k,j} = \sin \left(\frac{\pi k j}{N_x} \right), \quad k, j = 1, \dots, N_x - 1,$$

$$d_k^y = -\frac{4}{h_y^2} \sin^2 \left(\frac{k\pi}{2N_y} \right), \quad k = 1, \dots, N_y - 1, \quad (\mathbf{P}_y)_{k,j} = \sin \left(\frac{\pi k j}{N_y} \right), \quad k, j = 1, \dots, N_y - 1.$$

Introduce the array $\mathbf{H} = (h_{i,j})_{(N_x-1) \times (N_y-1)}$ with the element $h_{i,j}$ defined by

$$h_{i,j} = D \left(\frac{d_i^x + d_j^y + \frac{1}{12}(h_x^2 + h_y^2)d_i^x d_j^y}{1 + \frac{1}{12}(h_x^2 d_i^x + h_y^2 d_j^y)} \right) - \kappa$$

and the array $\tilde{\mathbf{H}} = (\tilde{h}_{i,j})_{(N_x-1) \times (N_y-1)}$ with the element $\tilde{h}_{i,j}$ defined by

$$\tilde{h}_{i,j} = \frac{1}{1 + \frac{1}{12}(h_x^2 d_i^x + h_y^2 d_j^y)}.$$

It is easy to verify that $h_{i,j} < 0$ since

$$\begin{aligned} d_i^x + d_j^y + \frac{1}{12}(h_x^2 + h_y^2)d_i^x d_j^y &\leq d_i^x + d_j^y - \frac{1}{3}d_i^x - \frac{1}{3}d_j^y \\ &= \frac{2}{3}(d_i^x + d_j^y) < 0 \end{aligned}$$

and

$$1 + \frac{1}{12}(h_x^2 d_i^x + h_y^2 d_j^y) = 1 - \frac{1}{3} \sin^2 \left(\frac{i\pi}{2N_x} \right) - \frac{1}{3} \sin^2 \left(\frac{j\pi}{2N_y} \right) \geq \frac{1}{3} > 0.$$

Let us define an operation“(e*)” by taking exponentials of an array element by element as

$$(e^*)^{\mathbf{H}} = (e^{h_{i,j}})_{(N_x-1) \times (N_y-1)},$$

and another operator “ $\odot$ ” for element by element multiplication between two arrays of same sizes as

$$(\mathbf{M} \odot \mathbf{L})_{i,j} = (\mathbf{L} \odot \mathbf{M})_{i,j} = (m_{i,j} l_{i,j})$$

where $\mathbf{M} = (m_{i,j})$, $\mathbf{L} = (l_{i,j})$. Then, a variation of constant formula by exponential time integration for the ODE system (9) leads to

$$\begin{aligned} \mathbf{U}(t_{n+1}) = & \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left[(e^*)^{\mathbf{H} \Delta t_n} \odot \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}(t_n) \right. \\ & - \int_0^{\Delta t_n} (e^*)^{\mathbf{H}(\Delta t_n - \tau)} \odot \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathcal{F}(\mathbf{U}(t_n + \tau)) d\tau \\ & \left. + \int_0^{\Delta t_n} (e^*)^{\mathbf{H}(\Delta t_n - \tau)} \odot \tilde{\mathbf{H}} \odot \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathcal{W}(t_n + \tau) d\tau \right]. \quad (10) \end{aligned}$$

Remark 1 We note that the derivation of the formula (10) is based on the facts that the domain Ω is rectangular and $\mathbf{A}$ and $\mathbf{B}$ are diagonalizable. It is not applicable to the problems with complex domain geometries or for which eigenvalue decompositions of $\mathbf{A}$ and $\mathbf{B}$ are impossible or difficult to obtain. For those problems, the classic ETD discretizations and Krylov subspace approximations for computing exponential matrix functions could be employed (see [12]).

Remark 2 In the case of Dirichlet boundary condition the operation $\mathbf{P}_x \mathbb{X} \mathbf{V}$ is equivalent to applying a Discrete Sine Transform (DST) [24] to each column of $\mathbf{V}$ and $\mathbf{P}_x^{-1} \mathbb{X} \mathbf{V}$ does an inverse DST,

$$\mathbf{P}_x \mathbb{X} \mathbf{V} = \text{DST}_{N_x-1}(\mathbf{V}), \quad \mathbf{P}_x^{-1} \mathbb{X} \mathbf{V} = \text{iDST}_{N_x-1}(\mathbf{V}).$$

Similarly, $\mathbf{P}_y \mathbb{Y} \mathbf{V}$ means applying a DST to each row of $\mathbf{V}$ and $\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{V}$ does an inverse DST,

$$\mathbf{P}_y \mathbb{Y} \mathbf{V} = \text{DST}_{N_y-1}(\mathbf{V}^T)^T, \quad \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{V} = \text{iDST}_{N_y-1}(\mathbf{V}^T)^T.$$

For $0 \leq \tau \leq \Delta t_n$, let

$$\begin{aligned} \mathbf{F}(\mathbf{U}(t_n + \tau)) &= (f_{i,j}(t_n + \tau))_{(N_x-1) \times (N_y-1)} = \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathcal{F}(\mathbf{U}(t_n + \tau)), \\ \mathbf{W}(t_n + \tau) &= (w_{i,j}(t_n + \tau))_{(N_x-1) \times (N_y-1)} = \tilde{\mathbf{H}} \odot \mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathcal{W}(t_n + \tau), \end{aligned}$$

and combine the terms from the stiff nonlinear part and the nonhomogeneous boundary condition to set

$$\mathbf{R}(t, \mathbf{U}(t)) = \mathbf{W}(t) - \mathbf{F}(\mathbf{U}(t)).$$

Then the Eq. (10) can be written in a simple compact form:

$$\mathbf{U}_{n+1} = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left[(e^*)^{\mathbf{H} \Delta t_n} \odot (\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n) + \mathbf{Q}^R \right], \quad (11)$$

where

$$\mathbf{Q}^R = (q_{i,j}^R)_{(N_x-1) \times (N_y-1)} = \int_0^{\Delta t_n} (e^*)^{\mathbf{H}(\Delta t_n - \tau)} \odot \mathbf{R}(t_n + \tau, \mathbf{U}(t_n + \tau)) d\tau. \quad (12)$$

2.3 Explicit Compact High-Order ETD Runge–Kutta Schemes

Explicit numerical schemes by using the ETD approach together with high-order Runge–Kutta approximations will be developed for full discretization of (11). The simplest polynomial to approximate to $\mathbf{R}(t_n + \tau, \mathbf{U}(t_n + \tau))$ is the constant polynomial $P_0(\tau)$

$$\mathbf{R}(t_n + \tau, \mathbf{U}(t_n + \tau)) \approx P_0(\tau) \equiv \mathbf{R}(t_n, \mathbf{U}(t_n)).$$

Let us define

$$\mathbf{S}_{0,0}(\Delta t_n) = \left(\alpha_{i,j}^{(0,0)}(\Delta t_n) \right)_{(N_x-1) \times (N_y-1)}, \quad (13)$$

where $\alpha_{i,j}^{(0,0)}(\Delta t_n) = \phi_0(\Delta t_n) := \int_0^{\Delta t_n} e^{h_{i,j}(\Delta t_n - \tau)} d\tau$. Then we may approximate $\mathbf{Q}^R$ by

$$\mathbf{Q}^R \approx \mathbf{Q}_{(0)}^R = \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{0,0}(\Delta t_n), \quad (14)$$

and obtain the first-order (in time) compact ETD scheme (cETD1) for computing $\mathbf{U}(t_{n+1})$ as

$$\mathbf{U}_{n+1} = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{Q}_{(0)}^R \right). \quad (15)$$

We refer to [20] for details.

In order to derive fully discrete high-order compact ETD schemes, we adopt here the Runge–Kutta approximations for time integration as proposed in [5, 14]. Let $\{a_s^{(r)} = s/r\}_{s=0}^r$ for a given integer $r \geq 1$. The following interpolation polynomial $P_r(\tau)$ of degree r is employed to approximate $\mathbf{R}(t_n + \tau, \mathbf{U}(t_n + \tau))$:

$$P_r(\tau) = \sum_{s=0}^r l_{r,s}(\tau) \hat{\mathbf{R}} \left(t_n + a_s^{(r)} \Delta t_n \right),$$

where $l_{r,s}(\tau)$ are the standard Lagrange basis functions associated with $\{a_s^{(r)} \Delta t_n\}_{s=0}^r$ and $\hat{\mathbf{R}}(t_n + a_s^{(r)} \Delta t_n)$ is the value or an approximated value of $\mathbf{R}(t_n + a_s^{(r)} \Delta t_n, \mathbf{U}(t_n + a_s^{(r)} \Delta t_n))$. Let $\theta = \frac{\tau}{\Delta t_n}$. We present below the polynomials that will be used for the Runge–Kutta approximations:

$$\begin{cases} P_1(\tau) = (1 - \theta) \mathbf{R}(t_n, \mathbf{U}_n) + \theta \mathbf{R}(t_{n+1}, \tilde{\mathbf{U}}(t_{n+1})), \\ \tilde{\mathbf{U}}(t_{n+1}) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{Q}_{(0)}^R \right); \end{cases} \quad (16)$$

$$\begin{cases} P_2(\tau) = (1 - 2\theta)(1 - \theta) \mathbf{R}(t_n, \mathbf{U}_n) + 4\theta(1 - \theta) \mathbf{R} \left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}} \left(t_n + \frac{\Delta t_n}{2} \right) \right) \\ \quad + \theta(2\theta - 1) \mathbf{R} \left(t_{n+1}, \tilde{\mathbf{U}}(t_{n+1}) \right), \\ \tilde{\tilde{\mathbf{U}}} \left(t_n + \frac{\Delta t_n}{2} \right) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n/2} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{0,0} \left(\frac{\Delta t_n}{2} \right) \right), \\ \tilde{\mathbf{U}}(t_{n+1}) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \right. \\ \quad \left. \left(2\mathbf{R} \left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}} \left(t_n + \frac{\Delta t_n}{2} \right) \right) - \mathbf{R}(t_n, \mathbf{U}_n) \right) \odot \mathbf{S}_{0,0}(\Delta t_n) \right); \end{cases} \quad (17)$$

$$\begin{cases}
\hat{P}_2(\tau) = (1 - 2\theta)(1 - \theta)\mathbf{R}(t_n, \mathbf{U}_n) + 2\theta(1 - \theta)\left(\mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}}^1\left(t_n + \frac{\Delta t_n}{2}\right)\right) \right. \\
\quad \left. + \mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}}^2\left(t_n + \frac{\Delta t_n}{2}\right)\right)\right) + \theta(2\theta - 1)\mathbf{R}(t_{n+1}, \tilde{\tilde{\mathbf{U}}}(t_{n+1})), \\
\tilde{\tilde{\mathbf{U}}}^1\left(t_n + \frac{\Delta t_n}{2}\right) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n/2} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{0,0}\left(\frac{\Delta t_n}{2}\right) \right), \\
\tilde{\tilde{\mathbf{U}}}^2\left(t_n + \frac{\Delta t_n}{2}\right) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n/2} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) \right. \\
\quad \left. + \mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}}^1\left(t_n + \frac{\Delta t_n}{2}\right)\right) \odot \mathbf{S}_{0,0}\left(\frac{\Delta t_n}{2}\right) \right), \\
\tilde{\tilde{\mathbf{U}}}(t_{n+1}) = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n/2} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \tilde{\tilde{\mathbf{U}}}^1\left(t_n + \frac{\Delta t_n}{2}\right) \right) + \right. \\
\quad \left. \left(2\mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\tilde{\mathbf{U}}}^2\left(t_n + \frac{\Delta t_n}{2}\right)\right) - \mathbf{R}(t_n, \mathbf{U}_n) \right) \odot \mathbf{S}_{0,0}\left(\frac{\Delta t_n}{2}\right) \right).
\end{cases} \quad (18)$$

Then, $\mathbf{Q}^R$ can be approximated by

$$\mathbf{Q}^R \approx \mathbf{Q}_{(r)}^R = \int_0^{\Delta t_n} (e^*)^{\mathbf{H}(\Delta t_n - \tau)} \odot P_r(\tau) d\tau \quad (19)$$

which has an accuracy of order $r + 1$. Define

$$\phi_{i,j}^s(\Delta t_n) = \frac{1}{(\Delta t_n)^s} \int_0^{\Delta t_n} \tau^s e^{h_{i,j}(\Delta t_n - \tau)} d\tau, \quad s \geq 0.$$

It is easy to check that

$$\begin{cases}
\phi_{i,j}^0(\Delta t_n) = -\frac{1}{h_{i,j}} (1 - e^{h_{i,j}\Delta t_n}), \quad \phi_{i,j}^1(\Delta t_n) = -\frac{1}{h_{i,j}} \left(1 - \frac{\phi_{i,j}^0(\Delta t_n)}{\Delta t_n} \right), \\
\phi_{i,j}^2(\Delta t_n) = -\frac{1}{h_{i,j}} \left(1 - \frac{2\phi_{i,j}^1(\Delta t_n)}{\Delta t_n} \right), & h_{i,j} \neq 0, \\
\phi_{i,j}^0(\Delta t_n) = \Delta t_n, \quad \phi_{i,j}^1(\Delta t_n) = \frac{\Delta t_n}{2}, \quad \phi_{i,j}^2(\Delta t_n) = \frac{\Delta t_n}{3}, & h_{i,j} = 0.
\end{cases} \quad (20)$$

For the approximation with $P_1(\tau)$, define

$$\mathbf{S}_{1,s}(\Delta t_n) = \left(\alpha_{i,j}^{(1,s)}(\Delta t_n) \right)_{(N_x-1) \times (N_y-1)}, \quad s = 0, 1, \quad (21)$$

where

$$\alpha_{i,j}^{(1,0)}(\Delta t_n) = \phi_{i,j}^0(\Delta t_n) - \phi_{i,j}^1(\Delta t_n), \quad \alpha_{i,j}^{(1,1)}(\Delta t_n) = \phi_{i,j}^1(\Delta t_n).$$

Then we get the second-order (in time) compact ETD Runge–Kutta scheme (cETDRK2) for computing $\mathbf{U}(t_{n+1})$ as

$$\mathbf{U}_{n+1} = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{Q}_{(1)}^R \right), \quad (22)$$

where

$$\mathbf{Q}_{(1)}^R = \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{1,0}(\Delta t_n) + \mathbf{R}(t_{n+1}, \tilde{\tilde{\mathbf{U}}}(t_{n+1})) \odot \mathbf{S}_{1,1}(\Delta t_n). \quad (23)$$

For the approximation with $P_2(\tau)$, define

$$\mathbf{S}_{2,s}(\Delta t_n) = \left(\alpha_{i,j}^{(2,s)}(\Delta t_n) \right)_{(N_x-1) \times (N_y-1)}, \quad s = 0, 1, 2. \quad (24)$$

where

$$\begin{aligned}\alpha_{i,j}^{(2,0)}(\Delta t_n) &= \phi_{i,j}^0(\Delta t_n) - 3\phi_{i,j}^1(\Delta t_n) + 2\phi_{i,j}^2(\Delta t_n), \\ \alpha_{i,j}^{(2,1)}(\Delta t_n) &= 4(\phi_{i,j}^1(\Delta t_n) - \phi_{i,j}^2(\Delta t_n)), \\ \alpha_{i,j}^{(2,2)}(\Delta t_n) &= -\phi_{i,j}^1(\Delta t_n) + 2\phi_{i,j}^2(\Delta t_n).\end{aligned}$$

Then we get the third-order (in time) compact ETD Runge–Kutta scheme (cETDRK3)

$$\mathbf{U}_{n+1} = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \mathbf{Q}_{(2)}^R \right), \quad (25)$$

where

$$\begin{aligned}\mathbf{Q}_{(2)}^R &= \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{2,0}(\Delta t_n) + \mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\mathbf{U}}\left(t_n + \frac{\Delta t_n}{2}\right)\right) \odot \mathbf{S}_{2,1}(\Delta t_n) \\ &\quad + \mathbf{R}\left(t_{n+1}, \tilde{\mathbf{U}}(t_{n+1})\right) \odot \mathbf{S}_{2,2}(\Delta t_n).\end{aligned} \quad (26)$$

With the approximation $\hat{P}_2(\tau)$, we arrive at the fourth-order (in time) compact ETD Runge–Kutta scheme (cETDRK4)

$$\mathbf{U}_{n+1} = \mathbf{P}_y \mathbb{Y} \mathbf{P}_x \mathbb{X} \left((e^*)^{\mathbf{H}\Delta t_n} \odot \left(\mathbf{P}_y^{-1} \mathbb{Y} \mathbf{P}_x^{-1} \mathbb{X} \mathbf{U}_n \right) + \hat{\mathbf{Q}}_{(2)}^R \right), \quad (27)$$

where

$$\begin{aligned}\hat{\mathbf{Q}}_{(2)}^R &= \mathbf{R}(t_n, \mathbf{U}_n) \odot \mathbf{S}_{2,0}(\Delta t_n) + \frac{1}{2} \left(\mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\mathbf{U}}^1\left(t_n + \frac{\Delta t_n}{2}\right)\right) \right. \\ &\quad \left. + \mathbf{R}\left(t_n + \frac{\Delta t_n}{2}, \tilde{\mathbf{U}}^2\left(t_n + \frac{\Delta t_n}{2}\right)\right) \odot \mathbf{S}_{2,1}(\Delta t_n) + \mathbf{R}\left(t_{n+1}, \tilde{\mathbf{U}}\left(t_{n+1}\right)\right) \odot \mathbf{S}_{2,2}(\Delta t_n) \right).\end{aligned} \quad (28)$$

The overall cost of the above proposed cETDRK schemes in two dimensions is $O(N^2 \log_2 N)$ per time step where $N = \max\{N_x, N_y\}$ by use of FFT-based fast calculation techniques, and the total required memory is obviously $O(N^2)$.

Remark 3 We note that, for very small $|h_{i,j}| \neq 0$, the evaluations of $\phi_{i,j}^k$ ($k = 0, 1, 2$) based on the formulas (20) could encounter numerical instabilities due to large cancellation errors [22]. In our implementation, we simply set $h_{i,j} = 0$ when $|h_{i,j}|$ is smaller than a pre-given tolerance (for example, 10^{-5}) to avoid this issue and such approach may slightly lose some accuracies. Another effective approach is to adopt the contour integral technique proposed in [10, 22].

Remark 4 It is easy to find that the cETDRK4 scheme is identical to the cETDRK3 scheme when $\mathbf{R}(t, \mathbf{U}(t)) = \mathbf{W}(t)$ (i.e., $\mathcal{F}(\mathbf{U}(t)) \equiv 0$), thus a reduced accuracy will happen in this situation due to the fact that $\int_0^{\Delta t_n} P_2(\tau) e^{h_{i,j}(\Delta t_n - \tau)} d\tau$ does not produce a fourth-order accurate approximation to $\int_0^{\Delta t_n} \mathbf{R}(\tau) e^{h_{i,j}(\Delta t_n - \tau)} d\tau$, unlike the Simpson's rule used in the classic fourth-order Runge–Kutta method. Consequently, it also implies the accuracy of the cETDRK4 scheme could degenerate for problems with nonhomogeneous Dirichlet boundary conditions.

2.4 The Case of Periodic Boundary Condition

The proposed cETDRK schemes can be easily applied to the problem with periodic boundary condition. If a periodic boundary condition is imposed, then we may denote the unknowns as

$$\mathbf{U} = (u_{i-1,j-1})_{N_x \times N_y} = \begin{pmatrix} u_{0,0} & u_{0,1} & \cdots & u_{0,N_y-1} \\ u_{1,0} & u_{1,1} & \cdots & u_{1,N_y-1} \\ \vdots & \vdots & \ddots & \vdots \\ u_{N_x-1,0} & u_{N_x-1,1} & \vdots & u_{N_x-1,N_y-1} \end{pmatrix}_{N_x \times N_y},$$

and set $\mathbf{A} = \frac{D}{h_x^2} \mathbf{G}_{N_x \times N_x}^P$ and $\mathbf{B} = \frac{D}{h_y^2} \mathbf{G}_{N_y \times N_y}^P$ with

$$\mathbf{G}_N^P = \begin{pmatrix} -2 & 1 & 0 & 0 & \cdots & 0 & 1 \\ 1 & -2 & 1 & 0 & \cdots & 0 & 0 \\ & & \ddots & \ddots & \ddots & & \\ 0 & 0 & \cdots & 0 & 1 & -2 & 1 \\ 1 & 0 & \cdots & 0 & 0 & 1 & -2 \end{pmatrix}_{N \times N}.$$

In addition, we have $\mathcal{W} = \mathbf{0}$ in this case. $\mathbf{A}$ and $\mathbf{B}$ are still diagonalizable and have the eigenvalue decompositions $\mathbf{A} = \mathbf{P}_x \tilde{\mathbf{D}}_x \mathbf{P}_x^{-1}$ and $\mathbf{B} = \mathbf{P}_y \tilde{\mathbf{D}}_y \mathbf{P}_y^{-1}$ where

$$\begin{aligned} d_k^x &= -\frac{4D}{h_x^2} \sin^2 \left(\frac{(k-1)\pi}{N_x} \right), \quad k = 1, \dots, N_x, \\ (\mathbf{P}_x)_{k,j} &= \exp \left(-i \frac{2\pi(k-1)(j-1)}{N_x} \right), \quad k, j = 1, \dots, N_x, \\ d_k^y &= -\frac{4D}{h_y^2} \sin^2 \left(\frac{(k-1)\pi}{N_y} \right), \quad k = 1, \dots, N_y, \\ (\mathbf{P}_y)_{k,j} &= \exp \left(-i \frac{2\pi(k-1)(j-1)}{N_y} \right), \quad k, j = 1, \dots, N_y. \end{aligned}$$

The corresponding cETDRK schemes for the problem (1) with the periodic boundary condition are then as same as (22), (25) and (27) for the Dirichlet boundary condition.

Remark 5 For the case of periodic boundary condition, the operation $\mathbf{P}_x \otimes \mathbf{V}$ means applying a Discrete Fourier Transform (DFT) [20, 24] to each column of $\mathbf{V}$ and $\mathbf{P}_x^{-1} \otimes \mathbf{V}$ does an inverse DFT,

$$\mathbf{P}_x \otimes \mathbf{V} = \text{DFT}_{N_x}(\mathbf{V}), \quad \mathbf{P}_x^{-1} \otimes \mathbf{V} = \text{iDFT}_{N_x}(\mathbf{V}).$$

Similarly, $\mathbf{P}_y \otimes \mathbf{V}$ means applying a DFT to each row of $\mathbf{V}$ and $\mathbf{P}_y^{-1} \otimes \mathbf{V}$ does an inverse DFT,

$$\mathbf{P}_y \otimes \mathbf{V} = \text{DFT}_{N_y}(\mathbf{V}^T)^T, \quad \mathbf{P}_y^{-1} \otimes \mathbf{V} = \text{iDFT}_{N_y}(\mathbf{V}^T)^T.$$

Thus we still can reduce the computational complexity of the proposed schemes in two dimension from $O(N^3)$ to $O(N^2 \log_2 N)$ per time step by FFT-based fast calculations.

3 Linear Stability Analysis

In this section, we investigate linear stability of the stabilized compact ETD Runge–Kutta schemes, i.e., we consider the following linear equation

$$u_t = \mathbf{L}u + \lambda u, \quad (29)$$

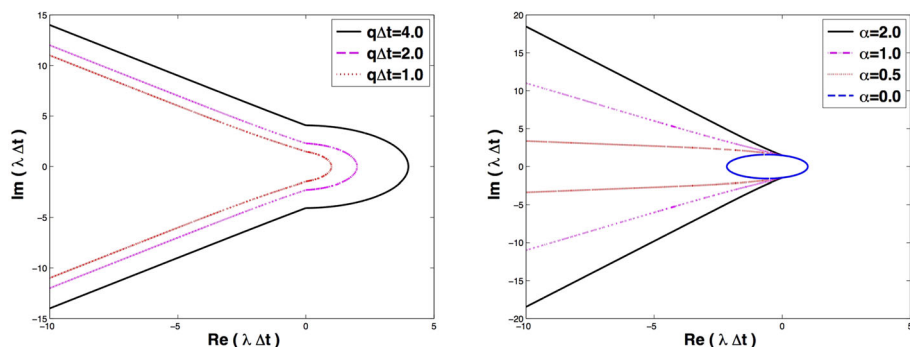


Fig. 1 Stability regions (the left part or interior of the stability boundary curves) of the stabilized first-order scheme (cETD1). Left $\kappa = \max\{0, -Re(\lambda)\}$ with $q\Delta t = 1.0, 2.0, 4.0$ respectively; right $q\Delta t = 1.0$ and $\kappa = \alpha \max\{0, -Re(\lambda)\}$ with $\alpha = 0, 0.5, 1.0, 2.0$ respectively

where $\mathbf{L}u = -qu$ with a homogeneous Dirichlet boundary condition, and study stability regions of the proposed schemes for (29). We note that compact representations of the spatial operator discretizations are mainly used to provide FFT-based fast calculation tools in practical implementation and have no effect on the numerical stability of the methods.

We follow the approach used in [10, 20] and consider the cases where λ is complex-valued and q is a positive real number that corresponds to a Fourier mode of the self-adjoint and elliptic operator $\mathbf{L}$. In the case of Dirichlet boundary condition, the proposed schemes are invertible. Next let us introduce a splitting parameter κ in adaptive way as

$$\kappa = \alpha \max\{0, -Re(\lambda)\}, \quad (30)$$

where $\alpha \geq 0$ is a constant to control the scale of the splitting.

For the stabilized first-order scheme (cETD1), we refer to [20] for its stability analysis, and present some of its stability regions in Fig. 1 for purpose of comparison. We also specially note that the stabilized first-order scheme has been shown to be unconditionally energy stable [37], i.e., in the sense the energy is monotonically non-increasing in discrete time regardless of the time stepsize.

For the stabilized second-order scheme (cETDRK2), let us rewrite its discretization for the model linear equation (1) as

$$u_{n+1} = (e^{(L-\kappa I)\Delta t} + (\lambda + \kappa I)\Delta t^{-1}(L - \kappa I)^{-2}(e^{(L-\kappa I)\Delta t} - I)^2 + (\lambda + \kappa I)^2\Delta t^{-1}(L - \kappa I)^{-3}(e^{(L-\kappa I)\Delta t} - I - (L - \kappa I)\Delta t)(e^{(L-\kappa I)\Delta t} - I))u_n, \quad (31)$$

where I is the identity operator. Clearly $\alpha = 0$ means that no splitting is applied. In order to satisfy the stability constraint (4), we need require $\alpha \geq 1/2$. In the case of $Re(\lambda) < 0$ we have the splitting parameter $\kappa = -\alpha Re(\lambda)$ according to (30). Set $x = Re(\lambda\Delta t)$, $y = Im(\lambda\Delta t)$. It is easy to get the amplification factor

$$\xi = e^{-(q\Delta t - \alpha x)} + ((1 - \alpha)x + yi) \frac{(e^{-(q\Delta t - \alpha x)} - 1)^2}{(q\Delta t - \alpha x)^2} + ((1 - \alpha)x + yi)^2 \frac{(e^{-(q\Delta t - \alpha x)} - 1 + (q\Delta t - \alpha x))(1 - e^{-(q\Delta t - \alpha x)})}{(q\Delta t - \alpha x)^3}. \quad (32)$$

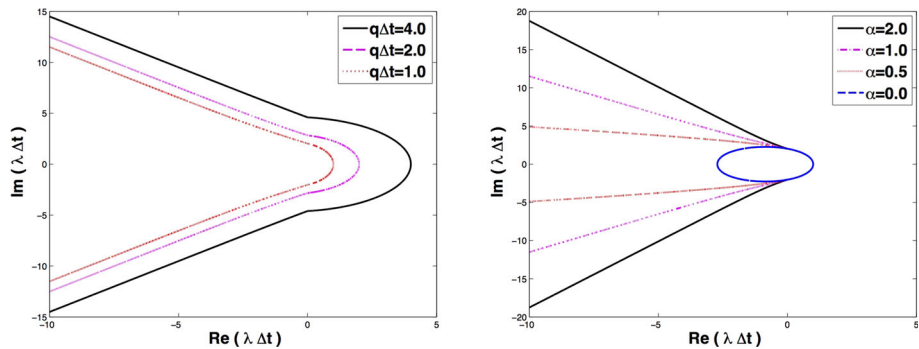


Fig. 2 Stability regions (the left part or interior of the stability boundary curves) of the stabilized second-order scheme (cETDRK2). Left $\kappa = \max\{0, -Re(\lambda)\}$ with $q\Delta t = 1.0, 2.0, 4.0$ respectively; right $q\Delta t = 1.0$ and $\kappa = \alpha \max\{0, -Re(\lambda)\}$ with $\alpha = 0, 0.5, 1.0, 2.0$ respectively

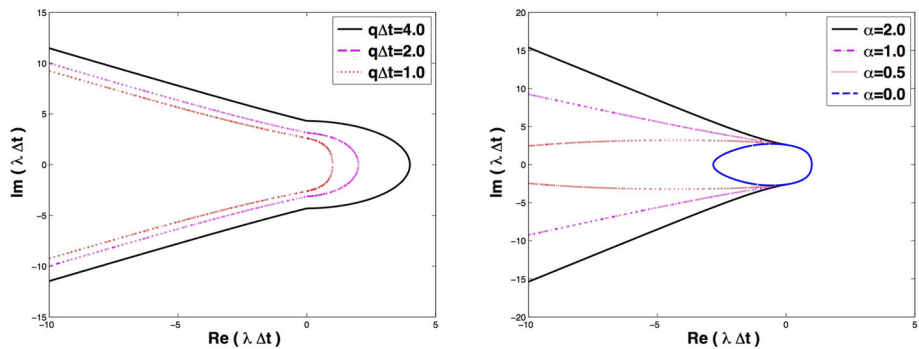


Fig. 3 Stability regions (the left part or interior of the stability boundary curves) of the stabilized third-order scheme (cETDRK3). Left $\kappa = \max\{0, -Re(\lambda)\}$ with $q\Delta t = 1.0, 2.0, 4.0$ respectively; right $q\Delta t = 1.0$ and $\kappa = \alpha \max\{0, -Re(\lambda)\}$ with $\alpha = 0, 0.5, 1.0, 2.0$ respectively

We then obtain the boundary locus curve equation of the stability region for the stabilized cETDRK2 in the left half plane ($Re(\lambda) < 0$) as

$$|\xi| = 1, \quad (33)$$

in which $Re(\lambda\Delta t)$ and $Im(\lambda\Delta t)$ are implicitly defined. Different values of $q\Delta t$ will give us a family of boundary curves for the corresponding stability regions. We note that in the case of $Re(\lambda) \geq 0$, it is easy to see that $\kappa = \alpha \cdot 0 = 0$ from (30) for the proposed schemes, and thus setting $\alpha = 0$ in (33) will give us the boundary curve of the stability region located for the stabilized cETDRK2 in the right half plane. Figure 2-left illustrates the stability regions of the stabilized second-order scheme with $\kappa = \max\{0, -Re(\lambda)\}$ for different values of $q\Delta t = 1.0, 2.0, 4.0$. Figure 2-right shows the stability regions of the second-order scheme with different values of $\kappa = \alpha \max\{0, -Re(\lambda)\}$ with $\alpha = 0, 0.5, 1.0, 2.0$, for a fixed $q\Delta t = 1.0$. Completely similar to the second-order case, we derive the boundary curve equations of the stability regions for the stabilized third-order (cETDRK3) and fourth-order (cETDRK4) schemes, and plot them in Figs. 3 and 4, respectively.

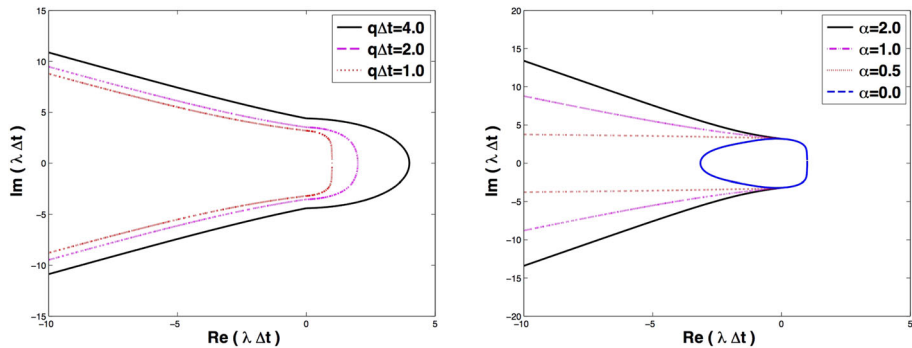


Fig. 4 Stability regions (the left part or interior of the stability boundary curves) of the stabilized fourth-order scheme (cETDRK4). Left $\kappa = \max\{0, -\text{Re}(\lambda)\}$ with $q\Delta t = 1.0, 2.0, 4.0$ respectively; right $q\Delta t = 1.0$ and $\kappa = \alpha \max\{0, -\text{Re}(\lambda)\}$ with $\alpha = 0, 0.5, 1.0, 2.0$ respectively

From Figs. 1, 2, 3, and 4, we observe that the stability region monotonically gets larger when the value of $q\Delta t$ gets bigger, and also monotonically grows larger along with the increasing of splitting parameter κ in all stabilized cETDRK schemes. We would like to specially note that larger κ may lead to a more stable but possibly less accurate scheme. It is also observed that the stability region only slightly shrinks along the increasing of the accuracy order for fixed q and κ .

4 Numerical Experiments

In this section, we will numerically demonstrate high-order accuracy and stability of the proposed compact ETD Runge–Kutta methods through various examples.

4.1 Linear Diffusion Equation

We first consider a linear diffusion equation with nonhomogeneous Dirichlet boundary condition.

Example 1 Let $\Omega = [-1, 1]^2$ and $T = 1$, we consider

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{1}{16\pi^2} \Delta u - \frac{1}{2}u, & (x, y) \in \Omega, t \in [0, T], \\ u(0, x, y) = \cos(2\pi x) \cos(2\pi y), & (x, y) \in \Omega. \end{cases} \quad (34)$$

The exact solution is set to be

$$u(t, x, y) = e^{-t} \cos(2\pi x) \cos(2\pi y), \quad (35)$$

which satisfies the equation and the initial condition. We test the case of Dirichlet boundary condition in which the boundary values are determined accordingly from the exact solution.

Since $f(u) = \frac{1}{2}u$ and $f'(u) = \frac{1}{2}$ in this example, we set $\kappa = \frac{1}{2}$. With such a splitting constant, it holds that $\hat{f}(u) = 0$ and then the cETDRK3 and cETDRK4 schemes are identical for this problem as stated in Remark 4. To test the spatial discretization accuracy of the proposed schemes, we took a fixed small time stepsize $\Delta t_n \equiv \Delta t = 1/2048$ (i.e., $N_t = 2048$), and uniformly refined the spatial grids (from a starting coarse grid of resolution $N_x \times N_y = 8^2$

Table 1 Errors and convergence rates of spatial discretization for Example 1 with *Dirichlet boundary condition* produced by the cETDRK3 scheme

$(N_x \times N_y) \times 2048$	8^2	16^2	32^2	64^2	128^2	256^2	512^2
<i>Accuracy test of spatial discretization</i>							
L^∞ error	4.8282e-03	3.7285e-04	2.4088e-05	1.5164e-06	9.4949e-08	5.9398e-09	3.7490e-10
CR	–	3.69	3.95	3.99	4.00	4.00	3.99

Table 2 Errors and convergence rates of temporal discretization for Example 1 with *Dirichlet boundary condition* produced by the cETDRK schemes

$(1024^2) \times N_t$	cETD1		cETDRK2		cETDRK3	
	L^∞ Error	CR	L^∞ Error	CR	L^∞ Error	CR
<i>Accuracy test of temporal discretization</i>						
4	1.0415e-01	–	1.7664e-03	–	2.5274e-05	–
8	4.8711e-02	1.10	4.7223e-04	1.90	3.0611e-06	3.05
16	2.3498e-02	1.05	1.2309e-04	1.94	3.7665e-07	3.02
32	1.1483e-02	1.03	3.1515e-05	1.97	4.6512e-08	3.02
64	5.6263e-03	1.03	7.9758e-06	1.98	5.7779e-09	3.01

to the finest level of 512^2) and repeatedly solved the Eq. (1) using the cETDRK3 scheme. To test the temporal discretization accuracy, we instead fixed a fine spatial grid ($N_x \times N_y = 1024^2$), then repeatedly refined the time stepsize (from initially $N_t = 4$ to $N_t = 64$ in the end) and solved (34) with the cETDRK schemes. Numerical results on the L^∞ errors at the final time $t = T = 1$ and corresponding convergence rates for Example 1 are reported in Tables 1 and 2. The errors are measured in the discrete L^∞ sense by taking the maximum absolute difference between the exact solution and the approximate solutions on all grid points. Clearly the exact solution (35) is very smooth. As expected we observe almost perfect fourth-order convergence for spatial discretization, and first, second, third convergence in time for the corresponding cETDRK schemes, respectively.

4.2 Allen–Cahn Equation

We next consider the Allen–Cahn equation (with a normalized diffusion coefficient) :

$$\frac{\partial u}{\partial t} = \Delta u - \frac{1}{\epsilon^2}(u^3 - u), \quad \mathbf{x} \in \Omega, t \in [t_0, t_0 + T], \quad (36)$$

where $\epsilon > 0$ often gives the length of the transition regions between the domains and u takes values between -1 and 1 . Note here $f(u) = \frac{1}{\epsilon^2}(u^3 - u)$, thus the Allen–Cahn equation has a very stiff nonlinear term when ϵ^2 is small. The Ginzburg–Landau free energy of the Allen–Cahn equation is then identified as

$$\mathcal{E}(u) = \int_{\Omega} \left\{ \frac{1}{2} |\nabla u|^2 + \frac{1}{4\epsilon^2} (u^2 - 1)^2 \right\} d\mathbf{x}. \quad (37)$$

Theoretically $|u| \leq 1$ and $\frac{f'(u)}{2} = \frac{1}{2\epsilon^2}(3u^2 - 1) \leq \frac{1}{\epsilon^2}$, thus we take $\kappa = \frac{2}{\epsilon^2}$ as a suitable splitting parameter which ensures (4).

4.2.1 Mean Curvature Flow

We consider the mean curvature flow problem governed by the Allen–Cahn equation.

Example 2 Let $\Omega = [-0.5, 0.5]^2$. The mean curvature flow problem we will simulate is given in the following:

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u - \frac{1}{\epsilon^2}(u^3 - u), & \mathbf{x} \in \Omega, t \in [0, T], \\ u(0, \mathbf{x}) = \tanh\left(\frac{R_0 - \|\mathbf{x}\|_2}{\sqrt{2}\epsilon}\right), & \mathbf{x} \in \Omega \end{cases} \quad (38)$$

with the periodic boundary condition being imposed. This is a typical example for testing performance of numerical solvers of the Allen–Cahn equation in many earlier works [7, 10, 37]. The above problem describes the shrinking process of a circle in two dimension. Let the radius of the circular region at time t be denoted by $R(t)$. As $\epsilon \rightarrow 0$, we know the theoretic limit radius $R_{lim}(t)$ satisfies

$$R_{lim}(t) = \sqrt{R_0^2 - 2t}.$$

We set $T = 0.075$ and the initial radius $R_0 = 0.4$, it is easy to calculate $R_{lim}(T) = 0.1$.

We chose three different values for ϵ as $\epsilon = 0.04, 0.02$, and 0.01 respectively, and numerically simulated the circle shrinking process for all cases. We focused on testing the temporal convergence of the proposed cETDRK schemes in this example, thus we used a fixed fine spatial grid of resolution $N_x \times N_y = 2048^2$ (i.e., $h_x = h_y = 1/2048$) for all simulations and uniformly refined time stepsizes starting from $N_t = 16$ to the finest level $N_t = 4096$. We found that all tests ran smoothly without any numerical instabilities. Numerical results on the approximate radii of the circle, errors and corresponding convergence rates for all cases at the final time $t = T = 0.075$ produced by the proposed cETDRK schemes are reported in Table 3. Since the analytic solution of Example 2 is unknown for any of the above given ϵ 's, we used the approximate results produced by the cETDRK4 scheme with the smallest time stepsize (i.e., $N_t = 4096$) as the benchmarks, i.e., we took $R_\epsilon \approx 0.09904566, 0.09962975, 0.10001512$ (See the bottom line of Table 3) for the cases of $\epsilon = 0.04, 0.02, 0.01$, respectively, and then used them to compute the errors of the simulated radii at the final time and corresponding temporal convergence rates. It is easy to see that the temporal convergences stably catch the expected optimal orders along the decreasing of the time stepsize for the case of $\epsilon = 0.04$. For the cases of $\epsilon = 0.02$ and $\epsilon = 0.01$ we observed that the temporal convergence rates monotonically get closer to the optimal ones although still not completely reached (especially in the case of $\epsilon = 0.01$). It basically implies that the smaller ϵ is, the finer time stepsizes are needed to achieve optimal convergence.

4.2.2 Phase Separation

We next consider numerical simulation of a process of phase separation (grain coarsening) described by the following example.

Table 3 Approximate radii of the shrinking circle, errors and convergence rates of temporal discretization at the final time $T = 0.075$ for Example 2 produced by the cETDRK schemes

$(2048^2) \times N_t$	$\epsilon = 0.04$			$\epsilon = 0.02$			$\epsilon = 0.01$		
	R	$ R - R_\epsilon $	CR	R	$ R - R_\epsilon $	CR	R	$ R - R_\epsilon $	CR
<i>Accuracy test of temporal discretization (cETD1)</i>									
16	0.375762	2.7672e-01	–	0.393992	2.9436e-01	–	0.398508	2.9849e-01	–
32	0.348105	2.4906e-01	0.15	0.387151	2.8752e-01	0.03	0.396831	2.9682e-01	0.01
64	0.303277	2.0423e-01	0.29	0.373133	2.7350e-01	0.07	0.393455	2.9344e-01	0.02
128	0.249826	1.5078e-01	0.44	0.345354	2.4572e-01	0.15	0.386615	2.8660e-01	0.03
256	0.200685	1.0164e-01	0.57	0.300329	2.0070e-01	0.29	0.372598	2.7258e-01	0.07
512	0.162690	6.3644e-02	0.68	0.246671	1.4704e-01	0.45	0.344814	2.4480e-01	0.16
1024	0.136502	3.7456e-02	0.76	0.197516	9.7887e-02	0.59	0.299763	1.9975e-01	0.29
2048	0.119953	2.0907e-02	0.84	0.159850	6.0220e-02	0.70	0.246087	1.4607e-01	0.45
4096	0.110235	1.1189e-02	0.90	0.134336	3.4706e-02	0.79	0.196972	9.6957e-02	0.59
<i>Accuracy test of temporal discretization (cETDRK2)</i>									
16	0.350486	2.5144e-01	–	0.387400	2.8777e-01	–	0.396846	2.9683e-01	–
32	0.301811	2.0277e-01	0.31	0.374117	2.7449e-01	0.07	0.393516	2.9350e-01	0.02
64	0.233115	1.3407e-01	0.60	0.347421	2.4811e-01	0.15	0.386861	2.8685e-01	0.03
128	0.168481	6.9435e-02	0.95	0.298925	1.9930e-01	0.32	0.373577	2.7356e-01	0.07
256	0.127120	2.8074e-02	1.31	0.230046	1.3042e-01	0.61	0.347201	2.4719e-01	0.15
512	0.108178	9.1322e-03	1.62	0.165652	6.6022e-02	0.98	0.298376	1.9836e-01	0.32
1024	0.101624	2.5781e-03	1.82	0.125423	2.5794e-02	1.36	0.229498	1.2948e-01	0.61
2048	0.099726	6.7989e-04	1.92	0.107792	8.1627e-03	1.67	0.165224	6.5208e-02	0.99
4096	0.099220	1.7413e-04	1.97	0.101907	2.2769e-03	1.84	0.125280	2.5265e-02	1.37

Table 3 continued

$(2048^2) \times N_t$	$\epsilon = 0.04$			$\epsilon = 0.02$			$\epsilon = 0.01$		
	R	$ R - R_\epsilon $	CR	R	$ R - R_\epsilon $	CR	R	$ R - R_\epsilon $	CR
Accuracy test of temporal discretization (cETDRK3)									
16	0.302846	2.0380e-01	—	0.374349	2.7472e-01	—	0.393532	2.9352e-01	—
32	0.228416	1.2937e-01	0.66	0.348506	2.4888e-01	0.14	0.386920	2.8690e-01	0.03
64	0.155094	5.6049e-02	1.21	0.299816	2.0019e-01	0.31	0.373797	2.7378e-01	0.07
128	0.114382	1.5336e-02	1.87	0.225197	1.2557e-01	0.67	0.347941	2.4793e-01	0.14
256	0.101877	2.8313e-03	2.44	0.152382	5.2753e-01	1.25	0.299230	1.9922e-01	0.32
512	0.099469	4.2351e-04	2.74	0.113444	1.3815e-02	1.93	0.224615	1.2460e-01	0.68
1024	0.099103	5.7719e-05	2.88	0.102126	2.4960e-03	2.47	0.151994	5.1978e-02	1.26
2048	0.099053	7.5296e-06	2.94	0.100001	3.7143e-04	2.75	0.113456	1.3441e-02	1.95
4096	0.099047	9.6007e-07	2.97	0.099680	5.0266e-05	2.89	0.102360	2.3449e-03	2.52
Accuracy test of temporal discretization (cETDRK4)									
16	0.251718	1.5267e-01	—	0.360848	2.6122e-01	—	0.390190	2.9017e-01	—
32	0.164032	6.4986e-02	1.23	0.321013	2.2138e-01	0.24	0.380212	2.8020e-01	0.05
64	0.112792	1.3746e-02	2.24	0.248112	1.4848e-01	0.58	0.360265	2.6025e-01	0.11
128	0.100522	1.4768e-03	3.22	0.160658	6.1028e-02	1.28	0.320385	2.2037e-01	0.24
256	0.099157	1.1145e-04	3.73	0.111768	1.2139e-02	2.33	0.247419	1.4740e-01	0.58
512	0.099005	7.3205e-06	3.93	0.100896	1.2659e-03	3.26	0.160102	6.0086e-02	1.29
1024	0.099046	4.5777e-07	4.00	0.099724	9.4321e-05	3.75	0.111759	1.1743e-02	2.35
2048	0.099045	2.6797e-08	4.09	0.099636	5.7967e-06	4.02	0.101152	1.1363e-03	3.37
4096	0.09904566	—	—	0.09962975	—	—	0.10001512	—	—

Table 4 Numerical errors and temporal convergence rates at the final time $T = 2048\delta$ ($\delta = 10^{-6}$) of Example 3 produced by the cETDRK2 and cETDRK4 schemes

Δt	64δ	32δ	16δ	8δ	4δ	2δ
<i>Accuracy test of temporal discretization (cETDRK2)</i>						
Energy error	7.922956e-01	4.136841e-01	1.967848e-01	8.191297e-02	2.850454e-02	8.649085e-03
CR	–	0.94	1.07	1.26	1.52	1.72
L^2 error	7.922956e-01	5.798729e-01	3.861934e-01	1.963603e-01	7.400435e-02	2.278182e-02
CR	–	0.35	0.59	0.97	1.41	1.70
<i>Accuracy test of temporal discretization (cETDRK4)</i>						
Energy error	2.556570e-01	9.035811e-02	1.849447e-02	2.209426e-03	1.820427e-04	1.171202e-05
CR	–	1.50	2.29	3.06	3.60	3.99
L^2 error	4.570676e-01	2.134216e-01	4.845723e-02	5.830356e-03	4.805087e-04	3.091941e-05
CR	–	1.10	2.14	3.05	3.60	3.99

Example 3 Let $\Omega = [-0.5, 0.5]^2$. The phase separation problem we will simulate is given by

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u - \frac{1}{\epsilon^2}(u^3 - u), & \mathbf{x} \in \Omega, t \in [0, T], \\ u(0, \mathbf{x}) = 0.2(2 \text{rand}() - 1), & \mathbf{x} \in \Omega. \end{cases} \quad (39)$$

where $\text{rand}()$ is a random number generator returning a real number between 0 and 1. The periodic boundary condition is again imposed.

We set the model parameter $\epsilon = 1/256$ for Example 3. A uniform space grid with $N_x = N_y = 2048$ (i.e., $h_x = h_y = 1/2048$) is again used to discretize the domain for all experiments in this example, and thus we have $h = \epsilon/8$ which is usually a fine enough resolution for spatial approximations for this problem [7, 37]. Set a time unit $\delta = 10^{-6}$. In order to avoid the effect of the randomness and discontinuity at the initial time, all convergence tests started with a same fixed initial configuration (see Fig. 5-left), which was obtained by running the cETDRK4 scheme until the stopping time $t = 128\delta$ with $\Delta t = \delta$ from a certain random initialization at $t = 0$. We then continued to run the cETDRK2 and cETDRK4 schemes with $\Delta t = 64\delta, 32\delta, \dots, \delta$ respectively, to a final time $t = T = 2048\delta$. Since the analytic solution of Example 3 is unknown, we took the numerical solution produced by the cETDRK4 scheme with the smallest time stepsize $\Delta t = \delta$ as the benchmark (approximate exact) solution. Numerical results on errors and corresponding temporal convergence rates at the final time are reported in Table 4. We can easily see from Table 4 that both errors measured by the energy difference and the L^2 -norm of the solution difference gradually achieve the second-order and the fourth-order convergence along the decreasing of time stepsize for the cETTK2 and cETDRK4 schemes, respectively. The plots of $\ln(\text{Energy})$ vs. $\ln(\text{Time})$ for both schemes are presented in Fig. 5-middle and Fig. 5-right for further illustration.

To further demonstrate stability of the proposed ETDRK schemes, we also set $\Delta t = 250\delta = 2.5 \times 10^{-4}$, and ran a simulation of the phase separation process from the initial time $t = 0$ to $t = 2048\Delta t$ by using the cETDRK4 scheme. Figure 6 presents the simulated phase configurations at some different times, from which we can observe some typical changes of the phase separation process modeled by (39). It is clear that the cETDRK4 scheme was

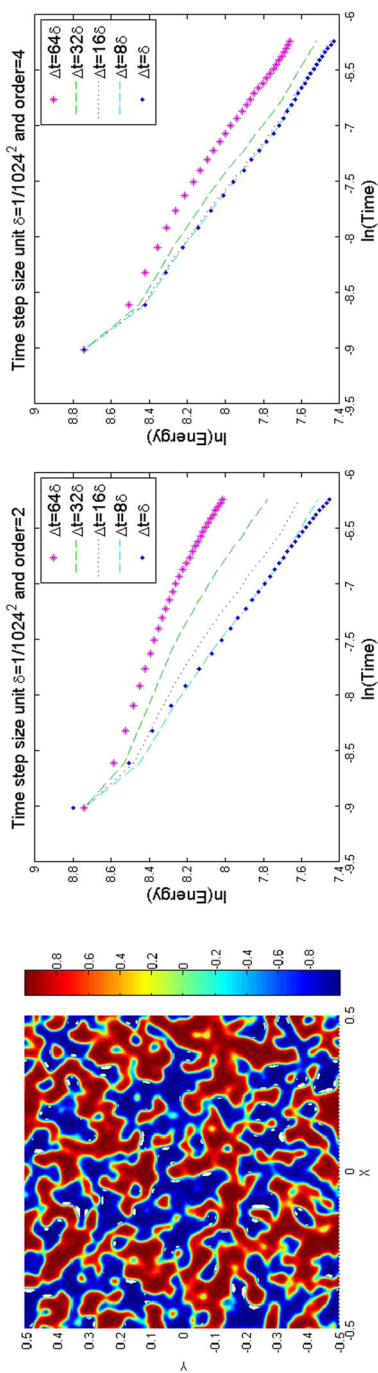


Fig. 5 Left the initial phase configuration; middle plots of $\ln(\text{Energy})$ versus $\ln(\text{Time})$ for the cETDRK2 scheme; right plots of $\ln(\text{Energy})$ versus $\ln(\text{Time})$ for the cETDRK4 scheme

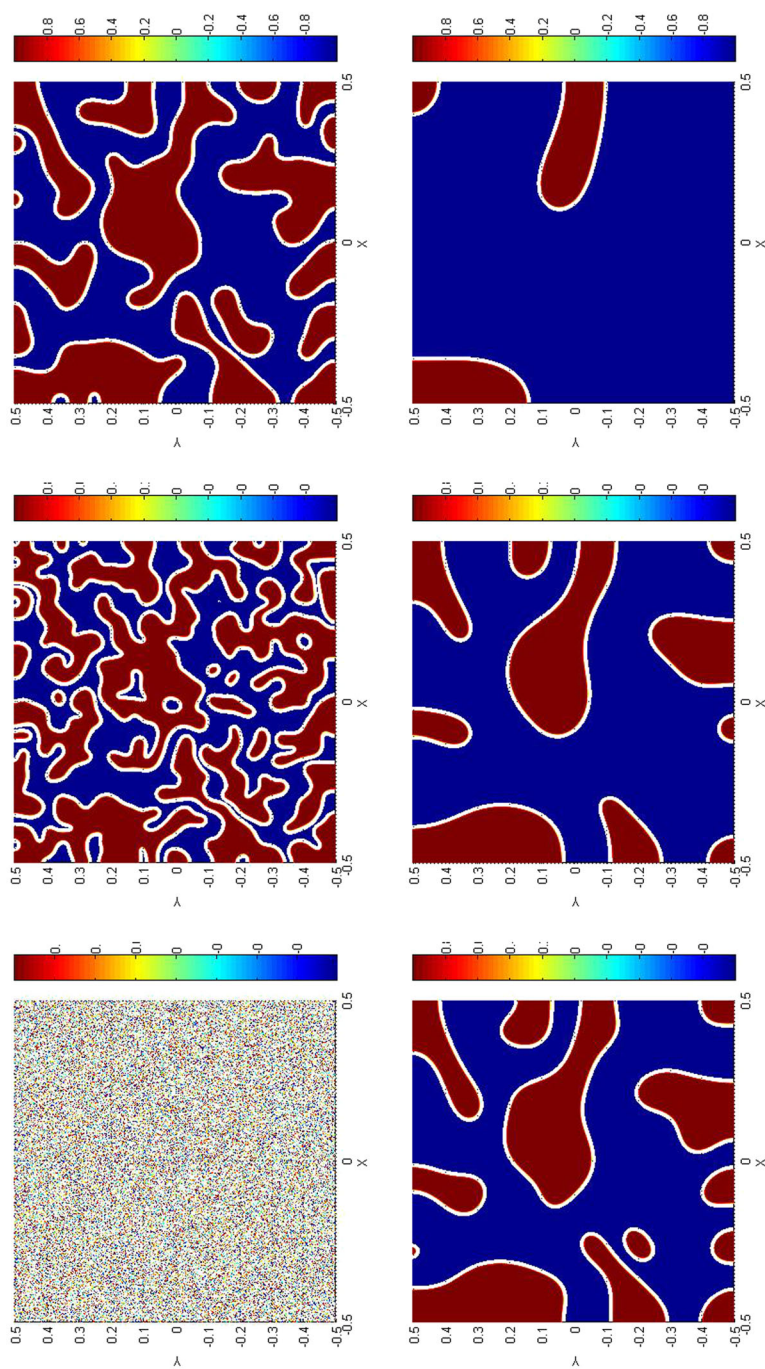


Fig. 6 Configurations of the simulated phase separation process at some different times by using the cETDRK4 scheme with $\Delta t = 2.5 \times 10^{-4}$. From *left to right* and *top to bottom*: $t = 0, 32\Delta t, 320\Delta t, 384\Delta t, 768\Delta t, 2048\Delta t, 6048\Delta t$

still numerically very stable and did a nice job in accurately catching the dynamics of the process.

5 Conclusions

In this paper we have proposed and studied fast high-order numerical methods for solving a family of second-order semilinear parabolic equations with Dirichlet or periodic boundary conditions in regular domains. The proposed schemes are based on explicit exponential time differencing Runge–Kutta approximations for high-order time integration and a two-step compact difference scheme for fourth-order spatial approximation, that allow use of FFT-based fast calculation techniques. In addition, a linear splitting scheme is also incorporated in the methods to further improve their numerical stability. Although our algorithms and examples are for two dimensional space, their extensions into higher dimensional spaces ($\mathbb{R}^d$ with $d \geq 3$) is straightforward and the overall computational cost will be $O(N^d \log_2 N)$ per time step. We also note that for the model problem with Neumann boundary condition, the applicability of our approach is still an open question, since first $\mathcal{W}$ in this case can not be simply determined from the value of $\partial u / \partial n$ on the boundary, and second the spatial approximation for the case of Neumann boundary condition may not achieve fourth-order accuracy. It also remains very interesting to further theoretically investigate energy stability of the proposed ETDRK schemes and develop adaptive time-stepping versions of the methods as studied in [6, 30, 33].

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